

1. A firm's costs are given:

Quantity	TFC	TVC	TC	AFC	AVC	AC	MC
0	\$100	0					
1	\$100	30					
2	\$100	50					
3	\$100	60					
4	\$100	72					
5	\$100	85					
6	\$100	110					
7	\$100	140					
8	\$100	180					
9	\$100	230					
10	\$100	290					

- Calculate the table
- Suppose market price is \$30, how much are total profits?
- Suppose market price is \$50, how much are total profits?

2. Fill the columns in the following table. What quantity should a profit maximizing firm produce? Verify your answer with marginal reasoning.

Q	TFC	TVC	MC	P=MR	TR	TC	Profit
0	\$20	\$ 0		\$ 22			
1	\$20	10		\$ 22			
2	\$20	15		\$ 22			
3	\$20	25		\$ 22			
4	\$20	40		\$ 22			
5	\$20	60		\$ 22			
6	\$20	90		\$ 22			

3. Calculate the costs in the table:

Q	TFC	TVC	TC	AFC	AVC	AC	MC
0	\$1000	0					
1	\$1000	10					
2	\$1000	18					
3	\$1000	24					
4	\$1000	32					
5	\$1000	42					

4. Profit analysis of a simple firm: Let assume that market has set \$15 unit price for the firm's product.

Q	TFC	TVC	MC	MR	TR	TC	Profit
0	\$10	0					
1	\$10	10					
2	\$10	15					
3	\$10	20					
4	\$10	30					
5	\$10	50					
6	\$10	80					

5. Complete the following table then draw the relevant curves from the data (fixed cost is \$200).

Total Product	Total Variable Costs	Total Costs	Average Fixed Cost	Average Variable Cost	Average Cost	Marginal Cost
0	0					
1	20					
2	38					
3	58					
4	64					
5	76					
6	93					
7	114					
8	139					

6. Given that,

$$C=10q - 4q^2 + q^3$$

Find out:

- AC & MC functions.
- Prove that MC reach its minimum point before AC.
- What is minimum AC & MC?
- Where is AC=MC ?

7. A cost function is given $TC=15x - 6x^2 + x^3$

Find out:

- AC & MC functions.
- Prove that MC reach its minimum point before AC.
- What is minimum AC & MC?
- Where is AC=MC ?

8. Consider the following example. Eryn currently works for a corporate law firm. She is considering opening her own legal practice, where she expects to earn \$200,000 per year once she establishes herself. To run her own firm, she would need an office and a law clerk. She has found the perfect office, which rents for \$50,000 per year. She could hire a law clerk for \$35,000 per year. If these figures are accurate, would Eryn's legal practice be profitable?